VF Asia Footwear Laboratory Testing Manual

Test Method for Oil Resistance	**Test method #: VF-I05** **REF #: TM-R- 014** **Page 1 of 2**
Date Issued: January 17, 2018 **Revision #: 1** **Reason for revision**: Reduce the thickness of solid material test sample from 6.35 mm to 4±0.5 mm for fuel 2,2,4-trimethylpentane (ISO 20344)	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To determine the resistance of soling material to swelling when in the presence of specific liquids.

REFERENCES: SATRA TM 63, FIA Test Method No. 319, ASTM D471-98, ISO 20344:2011,

EQUIPMENT AND MATERIAL:

1. Clicker press
2. Cutting Die : 25.4mm x 25.4 mm
3. Analytical balance
4. Weighted specimen holder
5. Acetone or isopropyl alcohol
6. Distilled water
7. Blotting paper/filter paper
8. ASTM Fuel B
9. ASTM IRM 903 Oil
10. 2,2,4-trimethylpentane, general purpose reagent (ISO 20344:2011)
11. Rubber and any type of polymer for oil resistant test

SAMPLE PREPARATION:

1. Thickness of test sample:
   - 6.35 mm: solid material for *ASTM Fuel B, ASTM IRM 903 Oil*
   - 4±0.5 mm: solid material for 2,2,4-trimethylpentane *(ISO 20344:2011)*
   - "as is" (full thickness): expanded materials.

   Note: Deviations from this thickness should be noted.
2. Cutting of specimens:
   - 25.4mm x 25.4mm: ASTM Fuel B and ASTM IRM 903 Oil
   - 16 ± 1 mm diameter: 2,2,4-trimethylpentane (ISO 20344).

REPETITIONS: Test three (3) samples per material**.**

TEST PROCEDURE:

Test should be carried out at the standard laboratory condition at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

Note: *ASTM Fuel B, ASTM IRM 903 Oil and ISO 20344:2011 are flammable material which should be handled carefully. Contact with the skin and inhalation must be avoided. Protective gloves, gas mask and safety glasses should be worn.*

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1. Measure and record the dry weight of each specimen in air before immersion. **W_1**
2. Dip the specimen in acetone and rinse with distilled water.
3. Measure and record the weight of the specimen in distilled water by attaching it to the weighted specimen holder. Measure and record the weight of the specimen holder in water. Calculate the weight of the specimen in water by subtracting the weight of the specimen holder in water from the weight of the specimen and specimen holder in water. **W_2**
4. Immerse the specimens in the desired liquid:
   - When testing with ASTM Fuel B and ASTM IRM 903 oil, immerse the specimens for 46 ± 0.5 hours at 23°C.
   - When testing with 2,2,4-trimethylpentane (ISO 20344), immerse the specimens for 22 ± 0.25 hours at 23°C.
5. Immediately after removal from the immersion medium, quickly dip the specimen in acetone or isopropyl alcohol and blot lightly with filter paper.
6. Measure the weight of specimen in air after immersion: **W_3**
7. Measure the weight of the specimen in distilled water after immersion, Follow procedure in step 3. **W_4**
8. Repeat steps 1- 7 for the other specimens.

CALCULATION:

$$\% \text{ volume change} = \frac{(W_3 - W_4) - (W_1 - W_2)}{(W_1 - W_2)} \times 100$$

Where: W_1 = weight of specimen in air before immersion (g)
W_2 = weight of specimen in distilled water before immersion (g)
W_3 = weight of specimen in air after immersion (g)
W_4 = weight of specimen in distilled water after immersion (g).

REPORT:

1. % volume change
2. Include material description, temperature, time of immersion, thickness of sample, immersion medium, and any deviations from this procedure